

Data 102, Fall 2026

Homework 2

Analyzing the Distribution of a Test Statistic

1. Suppose that you are studying the relationship between two different strings of data, $\{X_i\}_{i=1}^n$ and $\{Y_i\}_{i=1}^n$. Without loss of generality, assume that both vectors are mean zero.¹

To study the relationship between X and Y , you calculate their empirical correlation.

- Let $\vec{X} = [X_1, X_2, \dots, X_n]$ and $\vec{Y} = [Y_1, Y_2, \dots, Y_n]$.
- Let $\hat{\sigma}(\vec{X})^2 = \frac{1}{n} \sum_{i=1}^n X_i^2$ denote the empirical variance. Define $\hat{\sigma}(\vec{Y})^2$ similarly.
- Let $\hat{c}(\vec{X}, \vec{Y}) = \frac{1}{n} \sum_{j=1}^n X_j Y_j$ denote the empirical covariance.
- Let:

$$\hat{r}(\vec{X}, \vec{Y}) = \frac{\hat{c}(\vec{X}, \vec{Y})}{\hat{\sigma}(\vec{X})\hat{\sigma}(\vec{Y})}$$

denote the empirical correlation between the two vectors.

Suppose that you observed $\hat{r}(\vec{X}, \vec{Y}) = 0.1$. You want to test whether the observed correlation is statistically significant. Would the observed correlation be implausibly large (generate a small p -value), under a null model where the correlation is zero?

- (a) Show that the empirical correlation $\hat{r}(\vec{X}, \vec{Y}) = \cos(\Theta)$ where Θ is the angle between the observed vectors $\vec{X}$ and $\vec{Y}$.
- (b) Let $\hat{X}$ and $\hat{Y}$ denote the unit vectors pointing in the same direction as $\vec{X}$ and $\vec{Y}$. As a null, suppose that the directions $\hat{X}$ and $\hat{Y}$ are chosen independently, and uniformly, from the n -dimensional sphere.
 - i. Let's start with an example. Show that, if $\{X_i\}_{i=1}^n$ are drawn i.i.d. from a normal distribution with mean zero and variance σ_x^2 , and the entries $\{Y_i\}_{i=1}^n$ are drawn i.i.d. from a normal distribution with mean zero and variance σ_y^2 , then $\hat{X}$ and $\hat{Y}$ are drawn uniformly from the n -dimensional sphere.²

¹Otherwise, we would run all of the subsequent calculations using the centered vectors.

²Hint: show that the associated joint distributions are unchanged by rotations.

- ii. *Optional Challenge:* Show that, if we assume that $\{X_i\}_{i=1}^n$ are drawn i.i.d. and that their joint distribution is symmetric under rotations, then $X_i \sim \text{Normal}(0, \sigma_x^2)$ for some σ_x^2 .³
- iii. Show that the distributions of Θ , and thus $\hat{r}$, are unchanged if we instead assume that $\hat{X} = [1, 0, 0, \dots, 0]$, and that $\hat{Y}$ is drawn uniformly from the n -dimensional sphere. We will use this form for the null to simplify the problem.
- (c) Show that, if $\hat{X} = [1, 0, 0, \dots, 0]$ and $\Theta = \theta$, then $\hat{Y} = [\cos(\theta), \hat{y}_2, \hat{y}_3, \dots, \hat{y}_n]$ where, $\hat{y}_{2:n}$, are restricted to a sphere, in $n - 1$ dimensions, with radius $\sin(\theta)$.
- (d) Use your answer to part (c) to show that, under the null, $\Theta \in [0, \pi]$ is continuously distributed with density function:

$$f_{\Theta}(\theta) = \frac{1}{z_n} \sin(\theta)^{n-2}$$

where z_n is some constant chosen so that the distribution is normalized.

- (e) Comment on whether this result requires specific joint distributions of $\vec{X}$ and $\vec{Y}$ (e.g. the null is a simple hypothesis as in (b.i) and (b.ii)), or whether it holds for a whole collection of joint distributions (e.g. the null is a composite hypothesis).
- (f) Use the change of density formula to find the distribution of $\hat{r}(\vec{X}, \vec{Y}) = \cos(\Theta)$ under the null. Plot the resulting density functions, up to proportionality (we don't know z_n yet, so plot $f_{\hat{r}}(r)/f_{\hat{r}}(0)$), for an increasing series of n . Based on your plots, how should z_n behave as n increases?
- (g) We can work out the formula for z_n in two ways.
- i. *Geometry:* Argue, by equating probabilities to proportions, that, $f_{\Theta}(\theta)$ is equal to the ratio of the surface area of two spheres. Identify the spheres, and use the formula for the surface area of an n dimensional sphere, to solve for z_n .
- ii. *Calculus (Optional):* The argument above relies on the formula for the surface area of a sphere in n dimensions. So, unless you solved for the surface area directly, the geometric approach is not self-enclosed.

For a self-enclosed proof, we can use calculus. Express z_n in terms of the integrals $I_m = \int_{\theta=0}^{\pi} \sin(\theta)^m d\theta$.

Then, to find the value of I_m , integrate by parts. Set $u = \sin(\theta)^{m-1}$ and $dv = \sin(\theta)d\theta$, then substitute $\cos(\theta)^2 = 1 - \sin(\theta)^2$ to express I_m recursively in terms of I_m and I_{m-2} . Solve for I_0 and I_1 , then close the recursion for even and odd m separately.⁴

³Hint: Let $f_X(x)$ denote the density function for any individual sample. Let $g(x) = \log(f_X(x)/f_X(0))$. Then show that rotational symmetry requires $g(x_1) + g(x_2) = h(x_1^2 + x_2^2)$ for some function h , and for all $[x_1, x_2] \in \mathbb{R}^2$. Substitute $x_2 = 0$ to show that $g(x) = h(x^2)$. Then, let $u = x_1^2$ and $v = x_2^2$. Show that any function that satisfies $h(u) + h(v) = h(u + v)$ for all $[u, v]$ in the positive quadrant must be linear and must pass through the origin.

⁴It may help to use the double factorial, $k!! = k \times (k - 2) \times (k - 4) \times \dots$

- (h) Like most test statistics constructed from a string of i.i.d. random variables, the distribution of $\hat{r}(\vec{X}, \vec{Y})$ will concentrate, under the null, as n diverges. To show that it concentrates:
- First, plot the null distribution of Θ and $\hat{r}$ for a series of increasing n .
 - Show that, under the null, $\mathbb{E}[\Theta] = \pi/2$. Note: you do not have to integrate.
 - Show that $1 - \sin(\Theta) \geq 0$ and $= 0$ if and only if $\Theta = \pi/2$.
 - Show that, if $\Phi \in [0, \pi]$ is a random variable, then $\mathbb{E}[1 - \sin(\Phi)] = 0$ if and only if Φ is guaranteed to take on the value $\pi/2$.
 - Show that $\lim_{n \rightarrow \infty} \mathbb{E}[1 - \sin(\Theta)] = 0$. To evaluate the integrals needed, check your answers to part (g).
 - Use your answer to show that, as n diverges, the probability that $\Theta \in \pi/2 \pm \epsilon$ converges to 1 for any $\epsilon > 0$.⁵
 - It follows that Θ converges to $\pi/2$, in probability, as n diverges.⁶ Argue, using the continuity of $\cos(\theta)$ that, if Θ converges to $\pi/2$ in probability, then $\hat{r}(\vec{X}, \vec{Y})$ converges to 0 in probability.
- (i) To complete our test, we need a procedure for computing p -values from observed statistics.
- Evaluate the p -value of the observation $\hat{r}(\vec{X}, \vec{Y}) = 0.1$ for an increasing series of n . To evaluate your p -value, integrate either the density of $\hat{r}$ or Θ numerically. Decide whether to use a one-sided definition, $\Pr(\hat{r}(\vec{X}, \vec{Y}) \geq 0.1)$, or a two-sided definition, $\Pr(|\hat{r}(\vec{X}, \vec{Y})| \geq 0.1)$.
 - Does p increase or decrease as n increases? Make sure your answer is consistent with part (g).
 - What is the smallest value of n such that the observation is significant ($p \leq 0.05$)?

Distinguishing Distributions

2. Suppose that we can observe $\{X_i\}_{i=1}^n$, drawn i.i.d. from an unknown distribution. Suppose that the unknown distribution is either f_0 or f_1 . We will try to determine which distribution produced our data by running a hypothesis test. We will treat the hypothesis, $X_i \sim f_0$ as our null.
- (a) By Neyman-Pearson, the most sensitive (powerful) tests all use test statistics constructed as injective, monotonic functions of the likelihood ratio. Show that the

⁵Hint: as a model argument, check the arguments used to derive Chebyshev's inequality from Markov's inequality, then the use of Chebyshev's inequality to prove the weak law of large numbers.

⁶This is an interesting fact. In high dimension, two random vectors are almost always close to perpendicular if they are drawn with uniformly chosen directions.

log-likelihood ratio is:

$$\log(\text{LR}(\vec{x})) = \sum_{i=1}^n \log(f_1(x_i)/f_0(x_i)) = \sum_{i=1}^n \log(\text{LR}(x_i))$$

- (b) So, consider the test statistic, $s(\vec{x}) = \frac{1}{n} \log(\text{LR}(\vec{x}))$.
- Show that a test that uses s will be maximally sensitive.
 - Is large s evidence in favor of, or against, the null?
 - Show that $S = s(\vec{X})$ is the sample average of the collection of random variables $\{R_i = \text{LR}(X_i)\}_{i=1}^n$.
- (c) We can use these observations to work out the asymptotic distribution of the likelihood ratio when n is large.

Let $R_i = \log(\text{LR}(X_i))$, $\mathbb{E}[R_i|H_0] = \mu_0$, $\mathbb{E}[R_i|H_1] = \mu_1$, $\text{SD}[R_i|H_0] = \sigma_0$, and $\text{SD}[R_i|H_1] = \sigma_1$. Assume that all of the expected values and standard deviations are finite. Then, show that, under hypothesis k , S converges in probability to μ_k , with deviations on the scale $\sigma_k/\sqrt{n}$, and is asymptotically Gaussian, in the limit as $n \rightarrow \infty$.

- (d) The Kullback-Liebler divergence (KL divergence) between two distributions is defined:

$$D(p||q) = \mathbb{E}_{X \sim p}[\log(p(X)/q(X))]. \quad (1)$$

KL divergences are nonnegative, and equal zero if and only if the distributions are identical. They are often interpreted as a measure of the “distance” between two distributions. More accurately, they measure how distinct the distributions are. In general, $D(p||q) \neq D(q||p)$ so the KL divergence is not a true distance.

Show that $\mu_1 = D(f_1||f_0)$ and $\mu_0 = -D(f_0||f_1)$.

- (e) Let $s_*(\alpha)$ denote the decision threshold we compare S to when testing the null hypothesis f_0 at significance level α (e.g. with false positive rate α). Using your observations about the asymptotic distribution of the test statistic S , argue that, as n goes to infinity, $s_*(\alpha)$ must approach $\mu_0 = -D(f_0||f_1)$ from above, for any $\alpha > 0$. Thus, the KL divergence determines the asymptotic value of the decision threshold.
- (f) The KL divergence between f_0 and f_1 also determines the asymptotic rate at which the probability of a false negative converges to zero, as n diverges. The Chernoff-Stein lemma establishes that, if we adopt a likelihood ratio test, then the probability of a false negative converges to zero exponentially in n , with rate equal to $D(f_0||f_1)$:⁷

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log(\Pr(\text{decide for } H_0|H_1)) = -D(f_0||f_1). \quad (2)$$

The larger $D(f_0||f_1)$, the faster the probability of a false negative decays as a function of n , so the fewer samples are needed to distinguish the distributions with high sensitivity and significance.

⁷Note: you cannot derive this result directly from the CLT applied to S since it is a large deviation result.

Find $D(f_0||f_1)$ for the hypotheses:

- i. $H_0 : X_i \sim \text{Exponential}(\lambda)$, and $H_1 : X_i \sim \text{Exponential}(a\lambda)$ (see HW 1).
- ii. $H_0 : X_i \sim \text{Normal}(0, 1)$, and $H_1 : X_i \sim \text{Normal}(\mu, 1)$ (see lecture 4).
- (g) The KL divergence is asymmetric since our ability to distinguish two distributions depends on which distribution produces the data. Consider the hypotheses $H_0 : X_i = 0$, and $H_1 : X_i \sim \text{Bernoulli}(\epsilon)$ for some small $\epsilon > 0$. These two hypotheses should be difficult to distinguish when ϵ is small.
 - i. Describe the test procedure you should use to decide between H_0 and H_1 given a string $\{X_i\}_{i=1}^n$.⁸
 - ii. Suppose $\{X_i\}_{i=1}^n \sim H_1$ i.i.d. Imagine that you first see X_1 , then X_2 , then X_3 , and so on. Let N be the first sample where you reject H_0 . How is N distributed?
 - iii. Use your answer to find the exact probability of a false negative, as a function of n . Show that the probability of a false negative decays exponentially in n .
 - iv. Find $D(H_0||H_1)$ and check that your previous answer matches the statement of the Chernoff-Stein lemma.
 - v. Now, try exchanging the roles of H_1 and H_0 . Show that $D(H_1||H_0)$ is infinite. Try to make sense of this result by studying the probability that we declare for H_1 , if the data is actually drawn from H_0 , but we test treating H_1 as our null.

Bias in Police Stops

3. The following example is taken from [1, Ch. 6]:

A study of possible racial bias in police pedestrian stops was conducted in New York City in 2006. Each of $N = 2749$ officers was assigned a score z_i on the basis of their stop data, with large positive values of z_i being possible evidence of bias. In computing z_i , an ingenious two-stage logistic regression analysis was used to compensate for differences in the time, place, and context of the individual stops.

To see how these z -scores are computed, check the [original paper](#). We provide the data in a file `policez.csv` on DataHub.

We often assume each police officer acts **independently** of each other and therefore z_i 's are **independent and identically distributed** (i.i.d.). Does this assumption hold? We will see more later in the question.

For now, if we assume that this is true, we can use hypothesis testing on each police officer to determine if they showed racial bias when stopping pedestrians. Formally, for the i th officer in the dataset, we have

- **Null Hypothesis:** $z_i \sim \mathcal{N}(0, 1)$; i.e. the i th officer does not show racial bias

⁸You don't need any fancy machinery here. If you start a long derivation, back up and apply common sense.

- **Alternative Hypothesis:** z_i follows some other probability distribution; i.e. the i th officer shows some racial bias

Note that throughout this question, the word “bias” refers to police officers’ racial bias, rather than the statistical term.

- In the paper, the authors show that if the z_i ’s are i.i.d, they should be distributed according to $z_i \sim \mathcal{N}(0, 1)$. Let’s see if this is true: in one plot, make a normalized histogram (e.g. a histogram with total area equal to one) of the z -scores and a line plot of the pdf of the theoretical null $\mathcal{N}(0, 1)$. Does the theoretical null fit the data exactly? If not, describe how the data differ from the pdf of the theoretical null.
- Compute p -values $P_i = \Phi(-z_i)$ (where Φ is the standard normal CDF) and then apply the BH procedure with $\alpha = 0.2$. Plot the sorted p -values as well as the decision boundary. You should find that you make **201 discoveries**.
- Looking at the data, we can get a better fit to the distribution of z -scores if we use $\mathcal{N}(0.10, 1.40^2)$, called the empirical null (instead of the theoretical null from part (a)). Repeat parts (a) and (b), treating the empirical null as the null distribution. If you repeat part (b), should find that you make **only 5 discoveries**.
- In practice, when the assumptions of our hypothesis tests are violated, we can alleviate adverse effects by utilizing the empirical null instead of the theoretical null to generate our p -values. However, this leads to a change in the way we interpret our results. Consider the following questions, and respond in plain english:
 - In this study, the researcher argues in favor of using the empirical null to get around violated assumptions of each test (namely, that z_i are i.i.d). What might be a reasonable explanation as to why our z_i ’s fail to meet this assumption?
 - If we use the empirical null, is this study design able to uncover department-wide patterns in racial bias? Why or why not?

p -values, FDR and FWER

- The `adult.csv` file contains data from a random sample of the US adult population. It includes two numerical fields: **Age** and **Hours worked per week**. It also includes four categorical fields (which we have binarized for you): Gender, Education, Marriage status and whether the person’s income is greater than \$50,000. We will use this dataset to test the hypotheses of whether each of the categorical fields have any effect on the expectation of the numerical fields. For example, one test tests whether married individuals work significantly more or less than unmarried individuals.
 - Write a function `avg_difference_in_means` that takes as input two column names: `binary_col`, the name of a column with binary data, and `numerical_col`, the name of a column with numerical data. The function should compute the p -value for a test of the following hypothesis test:
 H_0 : There is no difference in the average value of `numerical_col` between the two groups specified in `binary_col`.

H_1 : The average value of `numerical_col` is different for the two groups specified in `binary_col`.

For example, the result of `avg_difference_in_means('Post HS?', 'Age')` should be a p -value for testing whether there is a significant difference in age between college-educated and non-college-educated adults. You should use a permutation test (i.e., an A/B test from Data 8) to compute your p -values, using at least 25,000 permutations to form your final null distribution. Using such a large number of permutations will stabilize the p -values so that random noise is unlikely to lead to differing results across the class. On Datahub, running the full loop of tests should take a couple minutes.

Hint: It might be useful to recall how to run the simulations to get the necessary p -values. The [Data 8 Textbook](#) walks through an example of running a permutation test.

Hint: To shuffle a single column of a dataframe in pandas, you can use code similar to the following line. Make sure you use the correct arguments to the [sample method](#)!

```
df['my_column'] = df['my_column'].sample(...).values
```

- (b) Use your function to compute eight p -values, one for each possible combination of categorical and numerical column.
- (c) Suppose we use a naive p -value threshold of 0.05 to make a decision for each hypothesis test. Given the p -values from above, for which tests do we reject the null hypothesis?

Gender and Age, 50K and Age, Gender and hours per week

- (d) Suppose we want to guarantee a Family-wise Error Rate (FWER) of 0.05. Given the p -values from above, for which tests do we reject the null hypothesis?
- (e) Suppose we want to guarantee a False Discovery Rate (FDR) of 0.05. Given the p -values from above, for which tests do we reject the null hypothesis?

Hint: Use the Benjamini-Hochberg algorithm.

- (f) How do the results from (d) and (e) compare? Explain how and why these results are different.

Hint: Recall how FWER and FDR are conceptually different.

- (g) Most variables don't always fit neatly into binary categories. As described earlier, we binarized these columns for you. Look at the original data in `adult_original.csv`. For one categorical column, give an example of how that variable could have been binarized differently, and how that might change the results from the earlier parts.

You aren't required to do any computation for this part: just explain how you might binarize one variable differently, and how that might change the results or your interpretation of them.

Risk Minimizing Estimators

5. It is standard practice, in statistics, to select procedures by minimizing an expected error or loss. Here we will focus on estimation problems (estimating an unknown parameter given data drawn from a related distribution) and will derive Bayesian estimators by minimizing an expected posterior loss.

Suppose that Θ is a random, unknown parameter and we observe data $X = x$, where x is drawn from a distribution that depends on Θ .

Let $\hat{\theta}(x)$ denote a function that estimates Θ given the observation $X = x$.

Show that:

- (a) To minimize the expected squared loss:

$$\mathbb{E}_{\Theta|X=x} \left[(\hat{\theta}(x) - \Theta)^2 \right] \quad (3)$$

we should set $\hat{\theta}(x)$ to the mean of the posterior, $\mathbb{E}[\Theta|X = x]$.

- (b) To minimize the expected absolute loss:

$$\mathbb{E}_{\Theta|X=x} \left[|\hat{\theta}(x) - \Theta| \right] \quad (4)$$

we should set $\hat{\theta}(x)$ to the median of the posterior.

- (c) Suppose, for simplicity, that Θ is a discrete random variable. Then, to minimize the expected 0-1 loss:

$$\mathbb{E}_{\Theta|X=x} \left[\mathbf{1}(\hat{\theta}(x) \neq \Theta) \right] = \Pr(\hat{\theta}(x) \neq \Theta|X = x) \quad (5)$$

we should set $\hat{\theta}(x)$ to the mode of the posterior (the maximum a posteriori estimator).⁹

References

- [1] Bradley Efron. *Large-scale inference: empirical Bayes methods for estimation, testing, and prediction*. Cambridge University Press, 2012.

⁹If you'd like, it is worth trying to extend this definition to the case when Θ is a continuously distributed random variable. First, show that $\Pr(\hat{\theta}(x) \neq \Theta|X = x) = 1$ for any $\hat{\theta}(x)$ if Θ is continuous, so the definition stated above cannot be used to select an estimator. Then, try finding, instead, the limiting estimator that minimizes $\Pr(\hat{\theta} \notin \Theta \pm \epsilon)$, in the limit as ϵ approaches zero.